

Lab03

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1.

```
library(readr)
library(dplyr)
```

Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

filter, lag

Os seguintes objetos são mascarados por 'package:base':

intersect, setdiff, setequal, union

```
airports <- read_csv(
  "airports.csv",
  col_select = c(IATA_CODE, CITY, STATE, LATITUDE, LONGITUDE)
)
```

Rows: 322 Columns: 5

```
-- Column specification -----
Delimiter: ","
chr (3): IATA_CODE, CITY, STATE
dbl (2): LATITUDE, LONGITUDE
```

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
airlines <- read_csv("airlines.csv")
```

Rows: 14 Columns: 2

-- Column specification -----

Delimiter: ","

chr (2): IATA_CODE, AIRLINE

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
aux <- data.frame(  
  DESTINATION_AIRPORT = character(),  
  total_atraso = numeric(),  
  quantidade = numeric()  
)  
  
ler_parte <- function(x, pos) {  
  x <- x[substr(x$DESTINATION_AIRPORT, 1, 1) != "1", ]  
  x <- na.omit(x)  
  
  resumo <- x %>%  
    group_by(DESTINATION_AIRPORT) %>%  
    summarise(  
      total_atraso = sum(ARRIVAL_DELAY),  
      quantidade = n()  
    )  
  
  aux <-<- rbind(aux, resumo)  
}  
  
read_csv_chunked(  
  "flights.csv",  
  col_types = cols_only(  
    DESTINATION_AIRPORT = col_character(),  
    ARRIVAL_DELAY = col_double()  
  ),  
  chunk_size = 1000000,  
  callback = SideEffectChunkCallback$new(ler_parte)  
)
```

NULL

```
flights <- aux %>%
  group_by(DESTINATION_AIRPORT) %>%
  summarise(
    total_atraso = sum(total_atraso),
    quantidade = sum(quantidade)
  ) %>%
  mutate(atraso_medio = total_atraso / quantidade)
```

2.

- a) A chave em `flights` é `DESTINATION_AIRPORT`.
- b) A chave em `airports` é `IATA_CODE`.
- c) Não aparecem aeroportos presentes em `flights` e ausentes em `airports`.

```
anti_join(
  flights,
  airports,
  by = c("DESTINATION_AIRPORT" = "IATA_CODE")
)
```

```
# A tibble: 0 x 4
#   i 4 variables: DESTINATION_AIRPORT <chr>, total_atraso <dbl>,
#   quantidade <int>, atraso_medio <dbl>
```

- d) A operação usada é `left_join`. Como as chaves possuem nomes diferentes, elas precisam ser indicadas no argumento `by`.
- e)

```
flights <- flights %>%
  left_join(
    airports %>% select(IATA_CODE, CITY, STATE),
    by = c("DESTINATION_AIRPORT" = "IATA_CODE")
  )
```

3.

```
airports %>%
  group_by(STATE) %>%
  summarise(aeroportos = n()) %>%
  arrange(desc(aeroportos))
```

```
# A tibble: 54 x 2
  STATE aeroportos
  <chr>         <int>
1 TX             24
2 CA             22
3 AK             19
4 FL             17
5 MI             15
6 NY             14
7 CO             10
8 MN              8
9 MT              8
10 NC             8
# i 44 more rows
```

4.

```
library(ggplot2)
library(maps)

dados_mapa <- flights %>%
  left_join(
    airports %>% select(IATA_CODE, LATITUDE, LONGITUDE),
    by = c("DESTINATION_AIRPORT" = "IATA_CODE")
  )

mapa_eua <- map_data("state")

ggplot() +
  geom_polygon(
    data = mapa_eua,
    aes(x = long, y = lat, group = group),
    fill = "white",
    color = "gray"
```

```

) +
geom_point(
  data = dados_mapa,
  aes(
    x = LONGITUDE,
    y = LATITUDE,
    color = atraso_medio
  ),
  size = 2
) +
labs(
  x = "Longitude",
  y = "Latitude",
  color = "Atraso médio"
) +
coord_fixed(1.3) +
theme_minimal()

```

Warning: Removed 3 rows containing missing values or values outside the scale range (`geom\_point()`).

